```
airodump-ng mon0 --write name_of_file
```

OR

```
airodump-ng wlan0mon --write name_of_file
```

You will require enough data (a minimum of 10,000) for cracking a Wi-Fi network using this method

- Cracking the Wi-Fi

Finally, with 10,000 data packets ready, we can start with the cracking process. Let's use the previously downloaded software aircrack-ng to crack the password. This command must run on a new terminal with the original data capture running in the background.

```
aircrack-ng name_of_file-01.cap
```

The program will prompt which Wi-Fi to crack. Choose the Wi-Fi, and the rest will be done. If the password is not strong enough, then it will be cracked easily. If not, the program will prompt for more packets. The program will retry again when there are 5000 more packets, and so on till it cracks the password or is asked to halt.

The key will be in this format-

`aa:bb:cc:dd:ee:ff`

Remove the colons for the true password of the network

`aabbccddeeff` is the password of this network

WPA-2 (and WPA)

As mentioned earlier, WPA-2 is the most secure encryption to date. The only two viable options are to guess the password or to trick the user into giving up the password.

- Guess the password

Guessing requires two things: Guesses and validation. It's a trial and error process where you make many guesses and check whether they are right. You can do that by manually entering the password one by one in the interface provided by the OS, which would be extremely slow and cumbersome. Even a script for this purpose would take a lot of time as it will communicate with the AP for every guess. An alternate (and faster) method of validating a guess is to use a 4-way handshake. All that is needed is to capture the packets transmitted when a valid client connects. If these packets (the 4-way handshake) are available, the passwords can be validated against it. There are a few different ways of guessing the password: